

Project # 25M065CN

MACC # 25-0441

TITLE: BEQ FC66-HP476-BB137 Chiller Repairs

ATTACHMENTS:

Chiller Piping Diagram

Loop Cleaning Procedures

GENERAL DESCRIPTION:

Replace Chillers at the following buildings:

1. HP476 -125-ton Chiller & 1 Chilled Water Pump
2. BB137-110-ton Chiller & 1 Chilled water Pump & VFD
3. FC66-2-110 Ton Chillers & 6 Chiller Water Pumps

******It is the responsibility of the contractor to replace pump/pumps on site of each new chiller installation and assure the pump/pumps will operate the new chiller with the proper GPM. If the existing pump does not meet the new chiller requirements it is the responsibility of the contractor to resize the pump with the proper style and GPM or the chiller. Verifying pump voltages is the responsibility of the contractor.**

SCOPE OF WORK:

The contractor shall provide all material, labor, equipment, and supervision required to accomplish the following:

Detailed Requirements and Specifications:

The contractor shall make site visits coordinated with PWD, Scotty Sholar at scotty.sholar.civ@usmc.mil or call 910-389-4418.

Chiller Specifications and Base of Design:

1. All chilled water piping and strainers shall be cleaned, that is not being replaced, prior to installation of new chiller. See attachment for hydronic pipe cleaning and flushing procedures.
2. Replace chilled water pumps and VFD on all chiller installs. Pumps must meet the manufacturers' specifications for each chiller.
3. All Chillers regardless of the heat exchanger are to be provided with a 10-year parts, labor, and refrigerant warranty. Chiller may also have shell and tube evaporator. Tubes shall be mechanically cleanable with removable heads. Water shall run through the shell of the evaporator. Refrigerant shall run through the tubes of the evaporator.
4. Chiller shall be equipped with a 20amp GFCI convenience outlet for maintenance to be performed. Chiller disconnects shall be replaced and the load side wiring shall be replaced. New disconnect shall meet the manufacturers' recommendations. Heat trace shall be protecting the

evaporator and all piping that is subject to freezing. The 20amp heat trace circuit shall be on a separate circuit. It is the responsibility of the contractor to verify circuit is in place and operable. If not, a new circuit shall be run.

5. Chiller shall include a single point power connection to a non-fused disconnect switch with through the door handle and compressor circuit breakers.
6. Chillers must utilize variable speed condenser fan motors to optimize chiller efficiency and maintain proper head pressure. Condenser fan motors shall have TEFC enclosures, IP54 protection with class F motor insulation with internal motor protection imbedded inside the windings.
7. Condenser coils shall be constructed of copper tubes with aluminum fins. Copper tubes must be constructed of internally enhanced seamless 0.016" thick copper with nominal diameter of 3/8". Aluminum fins must be manufactured in a sinusoidal shape with ripple edges to maximize heat transfer. Aluminum fins must be 0.005" thick. Coils must be coated with corrosion resistant, polyurethane based coating rated for 10,000 hrs. per ASTM B-117 with no degradation.
8. Chiller unit panels, structural members, control, and electrical boxes shall be constructed of galvanized steel with a polyester oven baked powder paint that meets a minimum of 7,000 hrs. salt spray test per ASTM B-117. Any other paint process that meets less than 7,000 hours shall not be accepted.
9. Chiller base frame must be constructed of structural welded C-channel steel and coated as described above. Formed galvanized steel bases shall not be accepted.
10. Condenser coils shall be protected by louvered coil guard and compressor housing shall have equivalent protection.
11. Chiller shall sit on seismic feet. And rubber pads installed on bottom of rails to elevate side rails to allow wash down of debris from under chiller.
12. Chiller must include thermal dispersion type flow switches. Paddle type will not be accepted. The flow shall be factory installed at the evaporator outlet.
13. Chiller must include phase loss with under/over voltage protection with LED indication of type of fault to guard against compressor motor burn out.
14. Emergency chiller connections: a set of emergency tees shall be installed before entering the chiller on supply and return lines. The tees shall meet the diameter of the pipe. A butterfly valve and a bolted cover plate shall be installed.
15. Control panels shall have access to microprocessor (controller) for viewing and set up, without interrupting machine operation. Chiller shall include a centrally located weatherproof control panel including DDC microprocessor controller. Control box shall be designed in accordance with NEMA 3R rating. Controllers and screens must be located within the panel and not exposed to sunlight. All wiring within the control panel shall be ran through slotted wiring duct with covers. Loose wires with zip ties will not be accepted.
16. Each chiller must include a minimum of two circuits. Multiple compressors on one refrigerant circuit shall not be accepted. Each circuit must consist of one individual compressor, replaceable core filter drier, moisture indicating sight glass, expansion valve, discharge and suction compressor ports, and liquid line service valve.
17. Controls shall be BACnet/MSTP, the controller shall be in English, and must monitor phase loss, over and under voltage, line, and load monitoring. Manual latch on critical faults and delayed restart on non-latched errors. Chiller shall be enabled on 45-degree F and above. Any control interface presently installed and operable on chiller shall be reinstalled and made to operate.

This includes hardware, wire start-stop, alarm, trouble, or integration to a BACnet/MSTP communication bus. A current control status summary will be attached.

18. Chiller manufacture must have a service organization with at least one service technician within 35 miles of Camp Lejeune, NC.
19. Chiller must comply with Buy America Act.
20. Piping diagrams shall be developed by the contractor based on the chiller selection.
 - 1.1 All piping shall be approved by PWD/OPS prior to installation. Prior to installation submit **shop drawings** for chiller piping station, supports, electrical, control layout and fabrication etc. (1) Proposed fabrication & assembly (2) Installation – fit & attachment details of materials and equipment) (3) Include drawings, diagrams, layouts, schematics, descriptive literature, illustrations, and similar materials furnished by the contractor to explain specific portions of the work required by contract.
 - 1.2 All piping 4” and below is to be copper and piping 5” and greater shall be a minimum of schedule 40 welded steel. All steel piping shall be primed and painted with a minimum of two (2) coats of alkyd enamel. All external piping shall have heat trace to prevent freezing. All pipes shall be insulated with a minimum of 2” of polyisocyanurate insulation. All insulation shall be covered with aluminum UV cover. All piping shall be supported with pipe stands and or welded supports.
 - 1.3 All piping shall be supported with pipe stands and or welded supports. All piping supports must be installed accordance with manufacturer standardization society / standard practice (MSS SP58 – 2018) (MSS SP69), or Designer to provide pipe support detail in contract drawings. **Recommend pipe stanchion saddle – type 37. (All thread rod or split ring hinged, or 2 bolts pipe clamps should NOT be allowed in PIPE STANCHION SUPPORTS**
21. Contractor shall add a bypass line that allows the chiller to be isolated from the rest of the chilled water piping loop for post-installation/pre-startup system flushing. (detailed in piping diagram)
22. Electrical Feed
 - 1.1 All electrical feeds and controls shall conform to the “NEC”. Contractor shall replace disconnects and liquid tight conduit and connectors. Existing wiring can be reused if it meets manufacturers’ recommendations.
 - 1.2 All electrical wiring sizes, voltage and amp draws of equipment is the responsibility of the contractor.
 - 1.3 All information provided in this scope of work is for reference only.
 - 1.4 Site visits shall be required for all bidding contractors.
23. Record Pre-installation water-flow readings and post-installation water-flow readings. These readings must be submitted with Maximo data.

DETAIL REQUIRMENTS AND SPECIFICATION:

1. Provide spec sheet for material and equipment used prior to start of work for government approval. Provide a copy of Maximo Equipment Inventory Sheet for each piece of equipment removed and installed.

SPECIAL SCHEDULING AND ACCESS:

1. Scheduled outages are to be scheduled through PWD. Outages shall be scheduled fourteen (14) days prior to equipment shut down. contact Scotty Sholar 910-389-4418.

SPECIAL CONDITIONS:

Contract shall be completed 300 days after award. Contractor shall ensure all employees and subcontractors have access and are fully vetted.

The contractor shall give Public Works the right to all HVAC equipment, including but not limited to chillers, AHU's, DOAS units, and any DDC controllers that are involved with the sequence of operation for the buildings HVAC System.

The contractor shall give Public Works the right to reclaim all refrigerant from any HVAC system before any equipment is to leave the work area. Any controls that are on existing chiller that need to be installed on new chiller can be removed and installed by the contractor. After Public Works remove any parts as listed above, the chiller will be the property of the contractor to dispose of offsite.

SUBMITTALS:

All equipment shall/must be approved. Submit all proposals to the ROICC and PWD for consideration. The contractor shall be required to submit work schedules to POC for approval. The contractor shall submit to the Contracting Officer the manufactures product specifications, complete with technical product data. The product specifications shall, as a minimum, provide technical details on the following items:

1. Manufacturer's name and product line.
2. Manufacturer's warranty.

HAZARDOUS MATERIAL:

NONE

ENVIROMENTAL:

NONE